

PRAVEEN MARK

Python Developer | Automation Engineer

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PROFESSIONAL SUMMARY:

Python Developer and Automation Engineer with experience designing and developing workflow automation tools, data-processing systems, backend applications, and engineering productivity solutions. Skilled in building Python-based automation utilities, CAD data processing workflows, system integration tools, ETL pipelines, reporting systems, and AI-assisted applications. Proven ability to identify manual processes and transform them into scalable automation solutions that improve efficiency and reduce operational effort. Currently expanding expertise in AI Automation, Retrieval-Augmented Generation (RAG), LangChain, CrewAI, Dify, and agent-based workflow systems.

CORE TECHNICAL SKILLS:

Programming & Development

Python • C# • SQL • JavaScript • C# • Object-Oriented Programming (OOP) • Git • GitHub • REST APIs

Automation & Workflow Systems

Python Automation • Workflow Automation • Task Automation • Batch Processing • ETL Pipelines • File Processing • Data Processing • Reporting Automation • Process Optimization • Multithreading • Concurrency • System Integration

Engineering Automation

PyCATIA • CATIA V5 • Autodesk VRED • Pixyz Studio • CAD Processing • Engineering Automation • Asset Optimization • Workflow Automation

Backend & Data Systems

Django • REST APIs • API Integration • MySQL • SQLite • Pandas • NumPy • Data Cleaning • Data Transformation • Reporting Systems • Power BI

AI & Intelligent Automation

Open AI Whisper • LangChain (*Learning*) • CrewAI (*Learning*) • Dify (*Learning*) • n8n (*Learning*) • RAG (*Learning*) • Agentic AI Concepts (*Learning*)

Additional Tools

Unity (Working Knowledge) • Blender

PROFESSIONAL EXPERIENCE:

XR Systems & Automation Engineer

Tech Mahindra Limited | Client: Mahindra & Mahindra Limited

Aug 2024 – Present

Supporting immersive engineering systems, automotive visualization environments, engineering review workflows, and automation solutions.

- Designed and developed Python-based automation tools to streamline engineering workflows, reduce manual effort, and improve operational efficiency.
- Built workflow automation solutions for engineering data processing, asset preparation, review documentation, and system integration.

- Developed custom applications using Python, Autodesk VRED APIs, and CAD automation frameworks to support large-scale engineering environments.
- Automated repetitive review, validation, and preparation tasks through custom software solutions and scripting frameworks.
- Created utilities for data processing, file management, reporting workflows, and engineering productivity improvement.
- Collaborated with cross-functional engineering teams to identify process bottlenecks and implement automation-driven solutions.

Key Engineering Automation Projects

CAD Workflow Automation Tool (Python • PyCATIA • CATIA V5)

- Developed a Python automation tool for CAD data cleanup, geometry processing, and engineering asset optimization.
- Automated surface joining, redundant entity removal, parameter cleanup, and batch processing workflows.
- Improved consistency and reduced manual engineering preparation effort.

Tracker-Based Vehicle Alignment System (Python • Autodesk VRED • HTC Vive Trackers)

- Developed a Python-based alignment utility that automated positioning and transformation workflows.
- Implemented tracker-driven synchronization logic and spatial calculation systems.
- Reduced manual setup effort and improved workflow efficiency.

Immersive Vehicle Sectioning Tool (Python • Autodesk VRED • HTC Vive)

- Developed a real-time geometry inspection system for dynamic sectioning and analysis of complex engineering assemblies.
- Implemented multi-axis clipping controls, free-form interaction, and visualization workflows.
- Improved review efficiency by simplifying internal component inspection.

AI Voice Review Assistant (Python • Whisper • Autodesk VRED)

- Built an AI-assisted review documentation system using Python and OpenAI Whisper.
- Automated speech-to-text transcription of review discussions and engineering observations.
- Generated searchable documentation from recorded review sessions.

Immersive Object Manipulation Tool (Python)

- Developed a self-service object management system enabling users to reposition, organize, and control assets independently.
- Implemented object selection, visibility control, movement controls, and workflow automation utilities.
- Reduced dependency on engineering support during review activities.

Engineering Data Transfer Tool (Python)

- Developed a high-performance file transfer utility with hardware-aware auto tuning and multithreaded processing.
 - Implemented resumable transfers, progress monitoring, performance optimization, and automated dataset management.
 - Improved handling of large engineering datasets and reduced manual transfer effort.
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IT Engineer – VR Systems & Automation Engineer

Tata Elxsi Limited | Client: Mahindra & Mahindra Limited

Dec 2023 – Jul 2024

- Managed VR infrastructure, deployment, validation, and support activities.
- Maintained runtime systems, trackers, controllers, and engineering review environments.
- Developed Python scripts supporting documentation, reporting, and workflow automation.
- Assisted engineering teams through automation-driven process improvements.
- Supported review sessions, demonstrations, diagnostics, and system troubleshooting.

Python Data Analyst & Backend Engineer

Worktable Private Limited – Singapore (Hybrid)

Nov 2019 – Aug 2021

- Developed ETL pipelines for ingestion, transformation, validation, and reporting workflows.
- Built backend applications and services using Django.
- Created REST API integrations and automation workflows.
- Developed web scraping and data extraction solutions.
- Built reporting dashboards using Power BI.
- Optimized SQL workflows and data-processing systems.

Projects: Inventory Management System • Pingo Mart Ecommerce

Technical Support Engineer

Tic-Fibernet – Chennai

Sep 2017 – Nov 2018

- Installation and configuration support.
- Networking, diagnostics, and troubleshooting activities.
- Documentation and customer support workflows.

Service Technician

Hi-Rev – Chennai

Jul 2015 – Feb 2017

- Vehicle diagnostics and maintenance activities.
- Technical inspections and troubleshooting.
- Workshop operations and service coordination.

EDUCATION

BSc (Hons) Computing Science

Coventry University – PSB Academy, Singapore

Mar 2022 – Jun 2023

Diploma in Engineering Technology

Auston Institute of Management, Singapore

Feb 2019 – Nov 2019

Diploma in Automobile Engineering

Tamil Nadu Advanced Technical Training Institute, India

Jun 2013 – May 2015
